25900522 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEMBG200 - BIOINSTRUMENTATION AND TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. Why is a laser used as the light source in confocal microscopy? [K] / [CO1]
2. Which cuvette is used in visible spectrophotometry [K] / [CO4]
3. What is preparative centrifugation? [K] / [CO2]
4. Which type of repeat sequence is present in a VNTR? [U] / [CO5]
5. What is the purpose of the excitation filter in a fluorescence microscope? [K] / [CO1]

[1x5 = 5]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

6. Explain how turbidity is related to microbial growth. [U] / [CO4]
7. Comment on Western blotting. [K] / [CO3]

8. Why can't live specimens be observed in SEM? [U] / [CO1]

9. Why do smaller proteins migrate faster in PAGE? [U] / [CO3]

10. Which biological molecules are separated using AGE? [K] / [CO3]

11. What is the role of immersion oil in microscopy, and why is it used?. [K] / [CO1]

12. State why temperature control is critical in Isoelectric Focusing. [E] / [CO3]

13. Summarise the steps of Southern blotting. [U] / [CO3]

14. What are VNTRs, and why are they important in DNA fingerprinting? [K] / [CO5]

15. Explain the principle of centrifugation. [U] / [CO2]

16. List two advantages of light microscopy. [An] / [CO1]

17. What type of samples might require nested PCR for detection? [A] / [CO5]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

18. Explain the steps involved in Northern blotting and its application in gene expression studies. [U] / [CO3]

19. Describe the optical components of a dark field microscope and their functions. [K] / [CO1]

20. Explain the causes and effects of chromatic aberration in lenses. [U] / [CO1]

21. Discuss the use of SDS-PAGE in clinical diagnostics. [A] / [CO3]

22. Describe the basic components of a UV spectrophotometer and how they work together to measure absorbance. [U] / [CO4]

23. Describe the working of a phase contrast microscope with the role of the annular diaphragm and phase plate. [U] / [CO1]

24. Explain the principle and methodology of RFLP. [K] / [CO5]

[6x5 = 30]

Part D

Long Essay Questions

Answer any **1** questions

Each question carries **15** marks

25. Explain transmittance and absorbance—define both, derive their relationship, and interpret their significance in practical colorimetric analysis. [U] / [CO4]

26. Describe the working principle of a Transmission Electron Microscope. Discuss its advantages and limitations. [U] / [CO1]

[15x1 = 15]